

AI Integration Research

1. AI-Powered Language Assistance

The project requires AI services for the following reasons:

- Translation
- Grammar clarification
- Creation of exercises

Research shows that REST-based AI services (like the APIs for NLP models) enable backend systems to integrate intelligent features while keeping the extra computing load on the main server minimal. (Jurafsky and Martin, 2023).

2. Adaptive Learning Algorithms

Research on adaptive learning has also shown:

- Estimating skills for each subject
- Adjustment of difficulty adapted to the response time and accuracy
- Systems for personalized recommendations

In the case of early-stage adaptive systems, it was sufficient to use rule-based adaptation combined with simple predictive models. (Brusilovsky and Millán, 2007).